

Atishaya Maharjan

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EDUCATION

B.Sc. Joint Honours in Computer Science & Mathematics

Expected May 2027

Minor in Statistics, University of Manitoba

Winnipeg, Canada

GPA: 3.77/4.5 | Dean's Honour List

Selected coursework: Advanced Algorithms (A), Automata Theory (A), Algebra II (A), Probability II (A)

Current coursework: Linear Algebra and Matrix Analysis (graduate cross-listed), Linear Models, ODEs

Planned Winter 2027: Optimization, Computability and Complexity, Graph Theory II

RESEARCH INTERESTS

Online and approximation algorithms, learning-augmented algorithms, decision-making under uncertainty, combinatorial optimization, computational geometry, and learning theory.

MANUSCRIPTS

Atishaya Maharjan, Anurag Murty Naredla, and Stéphane Durocher.

Geometric Algorithms and Hardness for Fuel-Efficient Collaborative Delivery. In preparation for conference submission, 2026.

Andrii Arman, **Atishaya Maharjan**, and Mahsa N. Shirazi.

A Generating-Function Framework for the HK-Property in Perfect m -ary Trees. Manuscript in preparation, 2026.

RESEARCH EXPERIENCE

Undergraduate Researcher – Algorithms & Computational Geometry

Mar. 2024 – Present

GADA Lab, University of Manitoba – with Dr. Durocher and Dr. Murty Naredla

Winnipeg, Canada

- Collaborative delivery with drones of different fuel rates: developed an $O(n^2)$ **algorithm** for the line version, improving the previous $O(n^3)$ algorithm (WALCOM 2025).
- Developed an $O(n^3)$ algorithm for the L_1 version, which also gives a $\sqrt{2}$ -**approximation** for L_2 ; worked with the team on an NP-hardness proof for L_2 (under internal review).
- Designed a divide-and-conquer heuristic for Hamiltonian paths in spatially embedded graphs; sped up a noncrossing spanning-tree library by about 30%.

Honours Researcher – Calibrated Risk Prediction

Sep. 2026 – Apr. 2027

University of Manitoba – with Dr. Henry, Dr. Sayari, and Dr. Salehkalaibar

Winnipeg, Canada

- Predicting pathogen risk in potato and canola fields with graph neural networks and AlphaEarth geospatial embeddings.
- Focus on calibration, sparse and imbalanced labels, and models that keep working in new regions.

Undergraduate Researcher – Combinatorics & Graph Theory

Jan. 2024 – Jul. 2025

MXML Lab, University of Manitoba – with Dr. Arman and Dr. Shirazi

Winnipeg, Canada

- Built a generating-function framework for the Hurlbert–Kamat property of perfect m -ary trees; proved coefficientwise leaf-root domination for all $m \geq 2$.
- Derived a closed form for their independence number; tested conjectures in SageMath across 1,000+ instances.

Undergraduate Researcher – Hypergraph Machine Learning

May 2025 – Aug. 2026

TerraByte Machine Learning Research Group, University of Manitoba

Winnipeg, Canada

- Generalized Graph Neural Diffusion to hypergraphs; implemented four variants in PyTorch and PyTorch Geometric.
- Built a reproducible MLflow pipeline tracking 25+ experiments across 5+ datasets.

AWARDS, SCHOLARSHIPS & PRESENTATIONS

S.U.R.E. Poster Competition – Third Place , Computational Science Category	Aug. 2024
• Presented “ <i>Minimizing Maximum Interference in 1-Dimensional Networks</i> ” at the University of Manitoba.	
International Undergraduate Student Scholarship <i>University of Manitoba</i>	
AccelR8 Prairie AI Hackathon – Winner	2026
Winnipeg Open Data Datathon – 1st of 15 teams	2026
CSSA Design & Craft Case Competition – 1st Place	2026
CSSA Capture the Flag – 1st of 41 teams (2026), 2nd of 25+ teams (2025)	
Attended CCCG 2025 (Canadian Conference on Computational Geometry), York University	2025

INDUSTRY EXPERIENCE

Senior Software Engineer	Mar. 2024 – Present
<i>LG Electronics</i>	<i>Englewood Cliffs, USA (Remote)</i>
• Shipped Next.js/Redux/TypeScript B2B features for 1,000+ vendors , contributing to a 25% engagement lift.	
• Designed analytics and tagging frameworks adopted by 6+ product teams for real-time measurement.	
• Built AI-powered global search with 35 ms lower latency, and checkout flows that drove a 15% conversion lift .	
Tech Lead	Jan. 2026 – Present
<i>Vibecore – GitHub-integrated code audit platform</i>	<i>Remote</i>
• Lead a small team building a code audit platform for founders who build with AI coding tools.	
• Automated CI/CD from GitHub Actions to AWS EC2; configured S3, DynamoDB, IAM, and EBS/EFS storage.	
• Backend services in Express with BullMQ job queues and Redis, run with Docker Compose behind Caddy.	
Software Engineer	Jun. 2025 – Jan. 2026
<i>Hoek Insights</i>	<i>Amsterdam, Netherlands (Remote)</i>
• Built a Python package for Three.js visualizations in Jupyter, cutting render time by 38% and bundle size by 25%.	
• Shipped a FastAPI/PostgreSQL MVP with 18+ endpoints and 10 product workflows.	
• Kept staging latency under 200 ms with 95% integration-test coverage.	
Software Engineer	Aug. 2019 – Apr. 2024
<i>Timeero</i>	<i>Phoenix, USA (Remote)</i>
• Led frontend development of a TypeScript/React/Mapbox mileage-tracking app with 1,000+ active users .	
• Shipped 15+ geospatial workflows for segment mapping, mileage capture, and field operations.	
• Improved time-capture accuracy by 30% through Mapbox-based mapping.	

TEACHING & LEADERSHIP

CS Help Centre Leader & Teaching Assistant <i>University of Manitoba</i>	Jan. 2025 – Apr. 2025
• Mentored 100+ students in first- to third-year CS courses; graded 150+ assignments for Analysis of Algorithms.	
Coding Kickoff Facilitator <i>University of Manitoba</i>	Feb. 2025 – Mar. 2025
• Facilitated introductory workshops on development environments and Git for 50+ first-year students.	
Events Executive <i>.devClub, University of Manitoba</i>	Aug. 2023 – Jul. 2024
• Co-led workshops, hackathons, review sessions, and community events for CS students.	

SELECTED TECHNICAL PROJECTS

Brainstorm  <i>Authorization-Aware RAG & Data Systems</i>	
• Role-aware retrieval system over Jira, PostgreSQL, Redis, and Pinecone with source grounding and access control.	
WinLocate  <i>Applied Data Engineering & Mapping</i>	2026
• City-comparison and geospatial decision-support platform built on Winnipeg’s open city data.	

TECHNICAL SKILLS

Research: algorithm design, NP-hardness reductions, generating functions, experimental mathematics, calibration
Programming: Python, C++, C, R, SQL, SageMath, TypeScript
Scientific Computing & ML: PyTorch, PyTorch Geometric, NumPy, SciPy, scikit-learn, pandas, MLflow
Tools: Git, Linux, Docker, GitHub Actions, Jupyter